

The Alberta Buck - Cash (DRAFT v0.1)

Printable Bearer Notes, Targeted Gifts, and Programmable Money

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For forty years cryptographers have tried to digitize physical cash and failed. Chaum's DigiCash (1990s) had the mathematics but went bankrupt for want of merchant adoption. Bitcoin gave us digital scarcity but has no native bearer-paper form – Casascius coins and BIP-38 paper wallets are aftermarket wrappers, not first-class instruments. Central bank digital currencies threaten the opposite: digital surveillance with no bearer privacy at all.

The Alberta Buck identity, wormhole, and reserve machinery – combined with one new primitive, a global commitment pool (Wormhole "Global Commitment Pool") – finally permit a *native* digital-physical bridge. A **BUCK Note** is a slip of paper carrying a QR-encoded opening of an on-chain commitment. The same cryptographic primitive instantiates three radically different financial instruments depending on the spend predicate baked into the commitment:

- **Bearer notes** (Mode A): anyone who scans the QR can deposit. Walking-around money.
- **Targeted notes** (Mode B): only a specific identity can deposit. Private gifts; cold-storage.
- **Programmable notes** (Mode C): time-locked, oracle-conditioned, multi-signature. Escrow, traveler's cheques, conditional payouts.

Demurrage – the same mechanism that drives BUCK circulation on chain – solves physical cash's oldest problem: lost notes. A BUCK Note ages in your pocket at the same rate a wallet balance does. After 50 years, a forgotten note's full value has returned to the Jubilee account and been redistributed to issuers. The 5-10% of physical currency that humanity loses every century is no longer a permanent monetary extraction – it is a temporary loan to the holder, repaid by demurrage.

Issuance is on chain and identity-bound; circulation is private and offline; deposit finality is cryptographic. Regulators retain selective-disclosure access to the issuance trail without observing day-to-day circulation. This is stronger compliance than physical cash provides today, and stronger privacy than any CBDC has ever proposed. (PDF, Text)

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DRAFT

1 The Cash Paradox

Physical cash is simultaneously irreplaceable and impossible to digitize.

It is irreplaceable because it does things no digital instrument has ever matched. A \$20 bill works without a network, a battery, a bank, an ID check, or a counterparty to your transaction. It clears instantly, finally, and without fees. It is the single most successful monetary instrument in human history – so successful that roughly half the value of US currency in circulation is held overseas as a global store of value, in defiance of every theoretical reason it shouldn't be.¹

It is impossible to digitize because every attempt has reproduced the surveillance properties of bank money instead of the privacy properties of cash. Modern card networks log every transaction. Bank wire transfers require both parties to be identified. Stablecoins record every transfer publicly on chain. Central bank digital currencies, where they exist or are being designed (China's e-CNY, Europe's digital euro proposals), are explicitly programmable surveillance tools – the central bank can see, restrict, expire, or geofence any holder's balance at will.²

The closest existing analogues to digital cash are aftermarket bearer wrappers around cryptocurrencies: Casascius physical Bitcoin coins (2011-2013), OpenDime USB sticks, BIP-38 paper wallets, and the various scratch-card "crypto gift cards" sold at convenience stores. These work, technically – a Casascius coin is a genuine bearer instrument – but they are workarounds, not designs. The underlying token (Bitcoin) does not natively understand bearer-paper form. Loss of the physical coin is permanent loss of the value. Counterfeiting requires faith in the manufacturer's hologram, not in the cryptography.³

What if the bearer-paper form were a *first-class* instrument of the system? Designed in from the start, with the same cryptographic and economic properties as on-chain balances, integrated with the identity and reserve mechanics that already exist?

That is BUCK Cash.

1.1 Why Forty Years of Trying

David Chaum invented digital cash in 1982 with the blind-signature e-cash scheme.⁴ By 1989 he had founded DigiCash to commercialize it. By 1998 DigiCash was bankrupt – not because the cryptography failed (it worked), but because no merchant wanted to be the first to accept it and no consumer wanted to be the first to hold it. The two-sided market never bootstrapped.

Stefan Brands extended Chaum's work in 1993 with offline divisible coins that revealed the spender's identity if (and only if) they double-spent.⁵ Camenisch, Hohenberger, and Lysyanskaya improved on this in 2005 with compact divisible e-cash.⁶ None of these schemes ever shipped in

¹Federal Reserve. "Currency in Circulation" (2024). Approximately half of US currency in circulation by value is held outside the United States, providing a global store-of-value function despite zero yield.

²Bank for International Settlements. "Rise of the central bank digital currencies" (2020). Surveys CBDC designs across major central banks; notes that nearly all incorporate explicit issuer-side surveillance and programmability features (spending limits, geofencing, expiration, freezing) that physical cash does not have.

³Caldwell, M. "Casascius Physical Bitcoins" (2011-2013). Physical brass and silver coins containing Bitcoin private keys under tamper-evident holographic seals. Production halted in 2013 after FinCEN classified the issuer as a money transmitter requiring federal licensing. Surviving "unpeeled" Casascius coins now trade at substantial premiums above their Bitcoin face value as collectibles.

⁴Chaum, D. "Blind Signatures for Untraceable Payments" (1982). *Advances in Cryptology - Crypto '82*, pp. 199-203. The foundational paper introducing blind signatures and digital cash. Chaum's DigiCash company commercialized the design from 1989-1998 before bankruptcy.

⁵Brands, S. "Untraceable Off-line Cash in Wallets with Observers" (1993). *Advances in Cryptology - Crypto '93*, pp. 302-318. Offline divisible e-cash with double-spend identity revelation – a deterrent rather than prevention model.

⁶Camenisch, J., Hohenberger, S., Lysyanskaya, A. "Compact E-Cash" (2005). *Advances in Cryptology - Eurocrypt*

production at scale.

Bitcoin, when it arrived in 2009, was deliberately *not* digital cash in the Chaumian sense. It was digital scarcity with public ledger transparency – closer to digital gold than digital cash. The bearer-paper market that grew up around it (Casascius, OpenDime, paper wallets) was an afterthought, not a design goal.

Zerocash (2014)⁷ and its production deployment as Zcash gave us the first scalable on-chain commitment-and-nullifier UTXO model – the cryptographic primitive that BUCK Notes use. Zcash showed that you can have private on-chain balances; it did not extend that to printable physical bearer instruments. Aztec extended the model with predicate-encrypted notes for private smart contract calls, but again, no paper.

The reason all of this never produced "BUCK Cash" – a printable, demurrage-aware, identity-aware bearer instrument – is that it required four pieces to come together simultaneously, and they did not until now:

1. A **commitment-and-nullifier UTXO model** on a smart contract platform (Zerocash gave us this in 2014; Aztec deployed it on Ethereum in 2018-2020).
2. A **cryptographic identity layer** that lets some notes be targeted to a specific recipient and others be left bearer (the BUCK identity system, with PS signatures, ElGamal credentials, and Chaum-Pedersen re-encryption).
3. A **demurrage mechanism** that solves the lost-cash problem and gives bearer notes a finite lifespan (BUCK's demurrage and Jubilee are unique among monetary systems).
4. A **jurisdictional reserve** that backs the value, so the bearer note represents a real claim on real assets, not a speculative crypto-token whose price might collapse between print and deposit (Alberta Buck's commodity-basket-stabilized reserve and the BuckK PID controller).

Alberta Buck has all four. No prior system had all four. That is why BUCK Cash is a *design*, not a workaround.

2 The Primitive: A BUCK Note

A BUCK Note is, physically, a slip of paper – or a card, or a steel plate, or a QR sticker on a fridge magnet. Cryptographically, it is a tuple:

(secret s , value v , age_at_print, t_{print} , predicate P , merkle_path)

These six values are encoded as a QR code (or NFC payload, or printed digits, or all three for redundancy). The slip of paper is the bearer instrument.

On chain, the BUCK contract maintains a Merkle tree of **note commitments**. Each commitment is computed as:

$$cm = H(s, v, age_0, t_{\text{print}}, P)$$

where H is a SNARK-friendly hash (Poseidon or Pedersen, per the wormhole construction). When a holder wants to deposit a note, they:

⁷05, pp. 302-321. Compact divisible e-cash where a single coin can be spent in pieces with $O(1)$ communication and storage per spend.

⁷Ben-Sasson, E., Chiesa, A., Garman, C., Green, M., Miers, I., Tromer, E., Virza, M. "Zerocash: Decentralized Anonymous Payments from Bitcoin" (2014). The foundational design for on-chain commitment-and-nullifier UTXO privacy. Deployed in production as Zcash since 2016.

1. Scan the QR code into their wallet app
2. The app constructs a SNARK proving:
 - Knowledge of $(s, v, age_0, t_{print}, P)$ opening some commitment in the tree
 - That the predicate P evaluates to true under the spender's witness
 - That the destination mint amount equals v less accumulated demurrage
3. The app submits the SNARK + a fresh nullifier $nf = \text{PRF}(s)$ to the chain
4. The contract verifies the proof, checks that nf is unspent, marks it spent, and mints to the spender's account

The contract sees only "valid SNARK + fresh nullifier + mint amount + recipient." It does not see which commitment was opened, who the original printer was, or what kind of note this is.

The note is now permanently dead. The paper is just a piece of paper. Tearing it in half is a **human** signal of finality, not a cryptographic one – but it's a useful ritual to prevent confused future holders from trying to redeposit a dead QR.

2.1 Counterfeiting is Cryptographically Impossible

A "fake" BUCK Note is just a random QR code. Its claimed commitment does not appear in the on-chain tree. The Merkle inclusion proof inside the SNARK fails. No mint occurs.

This is a stronger anti-counterfeit guarantee than the Bank of Canada provides for physical \$20 bills. Polymer banknotes can be counterfeited (the techniques are well-published; the question is only whether the counterfeit passes casual inspection). A BUCK Note cannot be counterfeited at all – the commitment registry is the authentic-bill list, and there is no inspection threshold to deceive.

Wallet apps that scan a BUCK Note display its status immediately:

Display	Meaning
VALID: 1,000 BUCK	Commitment in tree, nullifier unspent
DEAD	Commitment in tree, nullifier already spent
FAKE	Commitment not in tree
EXPIRED	Predicate includes an expiry that has passed
NOT FOR YOU	Predicate is targeted (Mode B), and you are not the recipient

Casual users do not need to understand the underlying cryptography any more than they need to understand the security features of a \$20 bill. They scan, see VALID, and accept the note as payment.

3 The Three Modes

The same primitive instantiates three radically different financial instruments by varying the spend predicate P .

3.1 Mode A: Bearer Notes ("Cash")

The simplest predicate:

$$P(\text{witness}) = \text{true if the witness reveals } s$$

Anyone who knows s can spend the note. This is exactly how a \$20 bill works: the note is bearer, possession is the entirety of ownership, no identity is checked, finality is immediate.

The cryptographic risk that physical cash does not have: anyone who **photographs** a bearer QR can spend it. No physical possession is required. A merchant's security camera that captures the back of a note while you tap-to-pay is a credible threat. This is solvable, but requires that bearer notes be printed under a **tamper-evident seal**:

- A peelable hologram covering the QR (the Casascius pattern – the act of peeling is the act of taking ownership)
- A scratch-off panel covering a separate one-time code that must be combined with the QR to spend (lottery-card pattern; lets the note be displayed without being live)
- A laminated sleeve that visibly damages on opening (mechanical seal, no electronic components)

A bearer note that has not yet had its seal broken is "in print but not yet alive." Once the seal is broken, the QR is exposed and the note becomes spendable by whoever holds it. This restores the physical-exclusivity property of paper cash.

The race condition that emerges if two people somehow both photograph an unsealed bearer QR is **exactly** the race that occurs if two people both find the same \$20 on the sidewalk: first to grab it (first to broadcast the spend transaction) owns it. The chain enforces atomic finality.

Use cases: tipping; small purchases; emergency money for travel; "kid's allowance" in physical form; merchant change-making; pocket-cash distributions in cash-poor communities.

3.2 Mode B: Targeted Notes ("Personal Cheques")

A more interesting predicate:

$P(\text{witness}) = \text{true}$ if the witness reveals s AND proves knowledge of sk for pk^*

where pk^* is a **stealth address** derived as

$$pk^* = H(s) \cdot pk_{\text{recipient}}$$

This binds the spendability to a specific identity without revealing whose identity it is to anyone reading the QR. Anyone can scan the note; only the recipient can prove knowledge of sk for pk^* (because $sk^* = H(s) \cdot sk_{\text{recipient}}$ and only the recipient knows $sk_{\text{recipient}}$).

The stealth address pattern is borrowed from Monero⁸ and Zcash shielded addresses. It guarantees that the note's targeting is **not observable on chain** until the recipient deposits. An attacker examining the commitment tree cannot tell which notes are bearer, which are targeted, or to whom the targeted ones are addressed.

The cryptographic machinery that makes this work – specifically, the Chaum-Pedersen proof of equality of discrete logarithms – is **exactly the same** machinery used in the BUCK Identity protocol's **approve** flow (Identity). Mode B targeting is not a new cryptographic component; it is a re-application of an existing one.

⁸van Saberhagen, N. "CryptoNote v 2.0" (2013). Introduced one-time stealth addresses for the recipient-untraceable transaction graph in Monero and related cryptocurrencies.

Use cases: gift cards; remittances to family abroad; private payments to specific counterparties; cold-storage savings targeted to your own future identity; estate planning (notes targeted to heirs, stored in a safe-deposit box); donations to specific charities without observable on-chain transfers.

3.2.1 Untraceable Targeted Transfer

The most novel use of Mode B: an untraceable transfer between two consenting identities.

Alice wants to send 1,000 BUCK to Bob without producing any observable on-chain link between them. She prints a Mode B note targeted to Bob's identity, hands him the paper (or sends him a photograph of the QR), and Bob deposits it.

The on-chain history shows:

- Alice's account decremented by 1,000 BUCK at the burn (issuance) step
- One commitment added to the tree
- Some time later, a SNARK consumes some commitment from the tree, mints to Bob's account

There is no link between Alice's burn and Bob's mint *except* the temporal correlation, which is dissolved by the size of the global anonymity set. If a thousand other people are also burning and minting around the same time, Alice's transfer is indistinguishable from any of theirs.

This is **better than physical cash** in one respect: physical cash transfers are traceable to the bank ATM where the cash was withdrawn, if the bank chooses to correlate serial numbers. BUCK Cash transfers are not traceable to the issuing account by anyone, including the issuer themselves, after the issuance step completes.

3.3 Mode C: Programmable Notes ("Smart Cheques")

The fully general case: P is an arbitrary boolean predicate over public state and private witness data.

$$P(\text{witness, public state}) = \text{any SNARK-encodable boolean function}$$

Examples worth singling out:

Predicate	Instrument
$\text{know}(s) \text{ AND } \text{own}(\text{pk}^*) \text{ AND } t > t_{\text{unlock}}$	Time-locked targeted note (savings)
$\text{know}(s) \text{ AND } \text{own}(\text{pk}^*) \text{ AND } t < t_{\text{expiry}}$	Expiring gift card
$\text{know}(s) \text{ AND } \text{multisig}(\text{pk}_1, \dots, \text{pk}_k \text{ of } n)$	Notarized escrow
$\text{know}(s) \text{ AND } \text{oracle}(\text{BTC_price} > X)$	Conditional payout / prediction voucher
$\text{know}(s) \text{ OR } (\text{own}(\text{pk}_{\text{issuer}}) \text{ AND } t > t_{\text{recovery}})$	Recoverable bearer note (issuer can recall)
$\text{know}(s) \text{ AND } (\text{own}(\text{pk}_a) \text{ OR } \text{own}(\text{pk}_b))$	Either-of-two recipients
$\text{know}(s) \text{ AND } \text{signature}(\text{pk}_{\text{authority}}, \text{transaction})$	Authority-cosigned transfer

The contract does not need to understand any of these. It sees only a SNARK proving "I opened a commitment in the tree and its predicate evaluates to true under my witness." The predicate logic is entirely inside the SNARK circuit, written by the note's printer at issuance time.

This is the substrate for BUCK Cash to extend beyond simple notes into a full family of programmable financial instruments, all sharing the same on-chain footprint and the same privacy properties.

Use cases: traveler's cheques (recoverable by issuer if lost, claimable by holder otherwise); inheritance instruments (claimable by heir after probate date); insurance payouts (claimable on oracle-confirmed event); escrow for real-estate closings; conditional bonuses (claimable if performance metric oracle-attested); payment streams (claimable in increments over time); subscription pre-pays (claimable by service provider in periodic increments).

4 Demurrage Solves the Lost-Cash Problem

This is the part of BUCK Cash that I think is genuinely novel, and I want to spend some time on it because it has implications well beyond the BUCK system.

4.1 The Hidden Inflation of Lost Cash

Every monetary system that uses physical bearer instruments suffers a constant slow loss to **unrecoverable currency**. Banknotes are lost in fires, dropped down storm drains, forgotten in the pockets of jeans that get washed and shredded, buried in landfills, sealed in walls, hidden by the elderly and forgotten by their heirs.

The Federal Reserve estimates that 5-10% of US paper currency is permanently unrecoverable at any given moment.⁹ At a US M0 of approximately \$2.3 trillion, this represents **\$115-230 billion in permanently extracted value** – seigniorage to the Federal Reserve, since the lost notes never get redeemed but the backing assets remain on the Fed's books.

This is not theft, but it is not neutral either. Lost cash is a regressive transfer: the people most likely to lose cash are the people who use cash most (the elderly, the unbanked, low-income workers paid in cash). The beneficiary is whoever issued the cash, generally the central bank, ultimately the government.

For private bank-issued cash (historical era; modern era in unusual cases like the Royal Bank of Scotland's banknotes still in circulation), the same dynamic applies to the issuing bank. Lost notes are pure profit.

4.2 What Demurrage Does

A BUCK Note carries two timestamps:

- age_0 : the BUCK-age the printer chose to bake into the note at issuance (typically 0 for "fresh", but could be positive to absorb a fractional remainder)
- t_{print} : the block timestamp at which the note was printed

When the note is deposited at time t_{spend} , the SNARK enforces:

$$v_{deposited} = v - 0.02 \times (age_0 + v \cdot (t_{spend} - t_{print}))$$

In English: the note's effective value decreases by 2% of its face value per year of elapsed time since printing. A 1,000 BUCK note printed today and deposited in five years deposits as 900 BUCK. In 25 years it deposits as 500 BUCK. In 50 years it is worthless.

⁹Federal Reserve Bank of San Francisco. "Diary of Consumer Payment Choice" (2018). Estimates 5-10% of US paper currency by value is permanently unrecoverable. At a US M0 of \$2.3 trillion, this represents \$115-230 billion in permanently extracted seigniorage.

The 50 BUCK of demurrage on a five-year-old note does not vanish. It flows to the **Jubilee account** described in Wormhole, which redistributes it pro rata to the issuers who pledged collateral to mint the original BUCK supply.

4.3 What This Means for Lost Notes

A BUCK Note that is lost in a fire, dropped down a storm drain, or hidden in a wall by a paranoid grandparent is **not** permanently extracted from the system. Its value decays to zero over 50 years, and during that decay, every BUCK of value lost to demurrage is automatically redistributed to BUCK issuers via the Jubilee account.

The 5-10% of currency that physical cash systems lose forever – BUCK Cash returns to the commons within a generation.

This has cascading consequences:

1. **Money supply self-correction.** No system-wide accounting tail of "missing" currency that the central authority must track and write off. Lost notes write themselves off automatically.
2. **Anti-hoarding incentive matches on-chain balances.** A BUCK held in your wallet or in your pocket as cash decays at the same rate. There is no advantage to converting on-chain BUCK to physical cash to evade demurrage; the demurrage clock runs the same way in both forms.
3. **Inheritance is automatic.** A grandparent who hides 100,000 BUCK in cash and forgets to tell anyone does not extract 100,000 BUCK from the economy on their death. Over the following 50 years, the value flows back to the Jubilee distribution – which (since BUCK issuers tend to be Albertan asset owners pledging real wealth) means it flows back, broadly, into the same economy where the grandparent originally earned it.
4. **No central bank profit from currency loss.** The seigniorage that accrues to central banks today from unrecoverable cash does not exist in BUCK Cash. Lost value does not benefit the issuer; it benefits the system as a whole.
5. **Counterfeit-recovery is automatic.** Even if a note printed years ago is somehow forged (which is cryptographically impossible, but humour the counterfactual), the value at stake is bounded by the demurrage decay. An attacker forging old notes would be forging worthless paper.

This is, I believe, the first monetary system in history where lost cash returns to the commons by automatic mechanism rather than by accounting fiat. It is a property that physical cash cannot have, because physical cash has no clock. It emerges naturally from BUCK's existing demurrage mechanism applied to the note-as-commitment primitive.

4.4 The Symmetry With On-Chain Balances

A second elegance worth pointing out: the demurrage rate on a BUCK Note is *the same* as the demurrage rate on an on-chain BUCK balance. This is by design, and it matters.

If physical BUCK Cash demurraged faster than on-chain BUCK, holders would prefer on-chain (and bearer cash would be discouraged, defeating the design's purpose). If on-chain BUCK demurraged faster than physical Cash, holders would race to print all their on-chain balances as physical notes (and the chain would lose its primary use as a settlement layer).

The matching demurrage rate makes the choice between on-chain and physical purely about **use case fit**: on-chain for routine commerce, automated payments, large transactions; physical for offline payments, gifts, savings under the mattress, disaster preparedness, and so on. Neither form is privileged by the demurrage mechanism.

5 Physical Security Model

The cryptography only protects what the cryptography touches. The physical paper itself needs a security model. This section sketches one.

5.1 Layered Defense

A printed BUCK Note has three concentric layers of security:

Layer	What it protects	Mechanism
Inner: secret	The cryptographic ability to spend	QR code containing s
Middle: physical seal	Bearer-mode exclusivity	Hologram, scratch-off, laminate sleeve
Outer: identity binding	Mode B targeting (only intended holder spends)	Stealth address $pk^* = H(s) \cdot pk_{\text{recipient}}$

The inner layer is the cryptographic core: as long as s is unknown, the note cannot be spent. This is unconditional protection against any threat that does not include observing s .

The middle layer is what makes bearer mode safe to circulate. An unsealed Mode A note is photographable, and once photographed by any observer, that observer can race to spend it. Sealing the QR under a tamper-evident covering means the note can be displayed (passed hand-to-hand, stored in a wallet, mailed in an envelope) without exposure. The seal must visibly damage on opening so that recipients can verify they were the first to expose the secret.

The outer layer is what makes Mode B notes safe to **publicly** distribute. A Mode B note can be photographed with no security implication: the photographer cannot spend it without also possessing $sk_{\text{recipient}}$. The targeting binding is cryptographic, not physical.

5.2 Recommended Manufacturing Profile

A practical BUCK Note specification, suitable for industrial-scale printing:

- **Substrate**: cotton-blend banknote paper or polymer (Bank of Canada-style). Steel or titanium plates for high-denomination "permanent" notes.
- **Printing**: standard inkjet or laser printing of QR codes is sufficient for cryptographic correctness. For visual anti-confusion (so people can tell a legitimate note from a printout), use UV-reactive ink, microprinting, or watermarks – the same techniques used for traditional banknotes. These do not add cryptographic security; they add **reassurance**.
- **Seal**: peelable hologram covering the secret QR (Mode A), or scratch-off panel covering a separate code (also Mode A), or no seal at all (Mode B and Mode C, since the cryptography handles exclusivity).
- **Redundancy**: print the QR twice (front and back) with Reed-Solomon error correction. A note that loses one QR to wear or damage can still be deposited via the second. For higher-value notes, also print the secret as base32 digits in human-readable form.

- **Issuer mark:** a non-cryptographic logo identifying the issuer (the Alberta Buck reserve, a credit union, a commercial issuer). This is for visual recognition only; the cryptographic authenticity check is the on-chain commitment.
- **Denomination indication:** visible face value (1, 5, 10, 100, 1,000, 10,000 BUCK) for human convenience. The actual value is what the on-chain commitment encodes; the printed face value is metadata. A wallet app that scans an apparent 1,000 BUCK note and reads "actual value: 100 BUCK" warns the user before deposit.

5.3 The Wallet App

The user-facing software is critical to the system's UX. A reasonable wallet app flow when scanning a BUCK Note:

1. Read the QR (or NFC payload, or human-typed digits)
2. Query the chain for the commitment's status:
 - In tree, nullifier unspent $\rightarrow$ VALID
 - In tree, nullifier spent $\rightarrow$ DEAD
 - Not in tree $\rightarrow$ FAKE
3. Check the predicate:
 - Mode A (bearer) $\rightarrow$ "Tap to Deposit"
 - Mode B (targeted) $\rightarrow$ if your stored identity matches pk^* , "Tap to Deposit"; else, "Not For You"
 - Mode C (programmable) $\rightarrow$ display the predicate in plain English, gate appropriately
4. On Deposit: construct the SNARK locally (or via a delegated prover service for thin clients), submit the transaction, display the result

This is a smartphone-class workload. SNARK construction for a single-note deposit is on the order of seconds on a modern phone (Aztec's Noir compiler and Zcash's Halo2 prover both run on mobile).

6 Use Cases

The same primitive serves a spectrum of use cases that today require entirely separate financial instruments.

6.1 Walking-Around Money

Print 100 x 10-BUCK bearer notes, fold them into your wallet, hand them out as cash. Recipients deposit when convenient or pass them on as change. The notes age at 2% per year, encouraging rapid circulation rather than hoarding.

This is the simplest use case and the closest analogue to a \$20 bill. The user experience is identical: hand someone paper, they have value.

6.2 Cross-Border Remittances

A BUCK Note is just paper. It crosses borders in pockets, envelopes, and parcels at zero marginal cost. Mode B (targeted to recipient's identity) makes the transfer useless to anyone except the intended recipient, so loss in transit is not a financial risk.

This compares favorably to the existing remittance industry, which extracts roughly 6.5% in fees globally on \$700+ billion in annual transfers.¹⁰ A BUCK Note remittance has a fixed cost: print + mail. The mail might cost \$1; print costs fractions of a cent; deposit gas costs a few dollars at most. At any reasonable remittance value, the fee compression is enormous.

The recipient does not need a smartphone or internet at the moment of receipt – they need them only at the moment they want to convert the note to on-chain BUCK or to local fiat. In countries with intermittent connectivity, this is a major usability improvement over digital remittance services.

6.3 Tip Jars and Charitable Donations

A printed sheet of small-denomination Mode A notes, perforated for tear-off, can be left next to a tip jar or a donation box. Patrons take notes (denominated at, say, 2 BUCK each), scan them with their phone, and the value flows to the tip jar's identity address.

This works without any intermediary, any payment processor fee, or any opening of an account. The patron pays in physical cash; the recipient receives in on-chain BUCK.

6.4 Cold Storage and Disaster Preparedness

A Mode C self-targeted note, time-locked until your future use, is a more durable form of cold storage than a 12-word seed phrase. The note is **self-evidently** an instrument of value – a customs officer, a relative, or your future self immediately recognizes a paper labeled "1,000 BUCK Note" as something worth keeping safe. A seed phrase, by contrast, is opaque to anyone who has not been trained to recognize it.

For disaster preparedness: a small folder of BUCK Notes in a fireproof safe survives infrastructure failures that would render exchange accounts and even hardware wallets unusable. As long as the chain is reachable somewhere in the world, the notes can be deposited. In a true grid-down emergency, the notes circulate hand-to-hand at face value (with the matching demurrage clock running uniformly, so prices remain stable in BUCK terms even during a multi-week outage).

6.5 Estate Planning

Mode B notes targeted to your heir's identity, stored in a safe-deposit box: zero probate process, zero tax-event delay, zero court intervention. The heir scans on your death, deposits, the BUCK is theirs.

For larger estates: Mode C notes with a predicate like "spendable by heir after death-certificate oracle-attested" enable conditional inheritance without a lawyer. The oracle could be a probate court, a notary network, or a death-registry integration – the SNARK does not care, as long as the predicate is well-defined.

¹⁰World Bank. "Remittance Prices Worldwide" (Q4 2024). Average global cost of remitting \$200 was approximately 6.5% in 2024. Total global remittance flows estimated at \$702 billion in 2024.

6.6 Gift Cards Without Issuer Risk

A traditional gift card carries the risk that the issuing merchant goes bankrupt between purchase and redemption (cumulative consumer losses to bankrupt gift-card issuers run into the hundreds of millions annually in North America¹¹). A Mode B BUCK Note targeted to "anyone" (Mode A) or to the recipient's identity (Mode B) carries no issuer risk: the value is on chain in the BUCK reserve, not on the merchant's books.

Specialized merchant gift cards become unnecessary – replaced by BUCK Notes with a small denomination, optionally bundled with a paper merchant receipt indicating the intended use.

6.7 Tax-Compliant Compensation

An employer paying a contractor in BUCK Notes faces the same tax-reporting obligations as paying in cash: payment is income, must be reported. The on-chain issuance trail makes this easier to comply with than physical cash, not harder – the employer's burn transaction is an auditable record of the payment, even though the contractor's deposit is private.

For employees: employers can issue Mode B notes targeted to each employee's registered identity, automating payroll while preserving employee privacy from external observers.

6.8 Underbanked and Cash-Economy Inclusion

This is, perhaps, the most important use case.

The Bank of Canada estimates 1-2% of Canadians are unbanked, with rates an order of magnitude higher in rural and Indigenous communities.¹² Globally the World Bank estimates 1.4 billion adults are unbanked.¹³

For these populations, physical cash is not a privacy preference; it is the only viable financial instrument. A digital-only currency (CBDC, stablecoin, traditional e-payment) excludes them by definition. BUCK Cash includes them: a person without a bank account, without a smartphone, without literacy in the local language, can hold and transfer BUCK Notes by physical handover, exactly as they hold and transfer paper currency today.

When and if they choose to interact with the digital economy, they deposit a note and the value moves on chain. Until then, they participate in the BUCK economy on exactly the terms they participate in the cash economy.

7 Selective Disclosure: Compliance Without Surveillance

Bearer instruments are politically controversial. Most jurisdictions either regulate them heavily (US bearer bonds were effectively abolished in 1982 by the Tax Equity and Fiscal Responsibility Act) or enforce strict reporting thresholds on large cash transactions (Canada's FINTRAC reports any cash transaction above \$10,000).

The concern is not bearer instruments per se; it is the **opacity to law enforcement and tax authorities** of large bearer-instrument flows. Money laundering, sanctions evasion, and tax evasion all exploit bearer-instrument anonymity.

¹¹Consumer Federation of America. "Gift Cards: A Costly Convenience" (2023). Estimates of cumulative consumer losses to bankrupt gift-card issuers in the United States range from \$200M-\$1B annually depending on methodology.

¹²Bank of Canada. "2022 Methods-of-Payment Survey" (2023). Approximately 1-2% of Canadians lack bank accounts; rates substantially higher in rural and Indigenous communities.

¹³World Bank. "The Global Findex Database 2021" (2022). An estimated 1.4 billion adults globally remain unbanked, with disproportionate concentrations in low-income countries, women, rural populations, and the elderly.

BUCK Cash threads this needle in a way that physical cash and CBDCs both fail.

7.1 The Asymmetry: Issuance vs. Circulation

Every BUCK Note's **issuance** is on chain and identity-bound. When Alice burns 5,000 BUCK from her registered identity to print five 1,000-BUCK Mode A notes, the chain records:

- Alice's identity (via the existing PS signature + ElGamal credential machinery)
- The amount burned
- The block timestamp
- Five new commitments added to the tree

This record is **permanent** and **cryptographically traceable to Alice's identity**.

The **circulation** of those notes – who holds them, who passes them on, who eventually deposits them – is private. The chain sees only the deposit transaction, which reveals only the recipient's address (the depositor) and the amount.

This produces an asymmetry:

- **Anyone** can see that "5,000 BUCK was issued as five Mode A notes by [some identity] at block 1,234,567"
- **Only Alice or a regulator with Alice's selective-disclosure key** can see that "the issuer was Alice"
- **Nobody** can see who held the notes between issuance and deposit
- **Anyone** can see "[some address] deposited a 1,000 BUCK Mode A note at block 1,234,890"

Linking issuance to deposit requires either: temporal correlation (defeated by a sufficiently large global note pool), or knowledge of s (which only a holder of the note has at any given time).

7.2 The Regulatory Audit Path

A regulator investigating illicit flows has a clean audit path:

1. Identify a suspect deposit on chain (via subpoena to the depositing party's identity, KYC obligations on the depositing wallet operator, etc.)
2. Subpoena the issuer of any BUCK held by the suspect: the **issuer** is identifiable from the on-chain issuance record (selective disclosure of the issuer's identity)
3. Examine the issuer's records for who they paid the note to (this is the issuer's responsibility, the same way a bank that issues a cashier's cheque is responsible for knowing who they issued it to)

The audit path stops at the first hand-to-hand transfer (which is intentionally private, as physical cash is private). But the **origination** of every note is traceable in a way that physical cash is not – a \$20 bill issued by the Bank of Canada is identical to every other \$20 bill, with no per-bill issuance trail at all.

7.3 What Compliance Looks Like

A reasonable regulatory regime around BUCK Cash:

- **Issuers** (anyone who burns on-chain BUCK to mint notes) must be identity-registered and may face reporting obligations on large issuances (analogous to current cash withdrawal reporting)
- **Depositors** (anyone who deposits a note to mint on-chain BUCK) face the same reporting obligations as cash deposits at a bank today
- **Holders** who simply pass notes hand-to-hand have no reporting obligation, the same way you have no obligation to report having \$200 in your wallet
- **Demurrage rate** makes large-scale long-term cash hoarding economically unviable (the 2%/year decay caps the value of cash-as-store-of-value at roughly the same level as a moderate-yield savings account loses to inflation)

This regime is **strictly more privacy-preserving** than the current Canadian dollar regime (which surveils every electronic transaction) and **strictly more compliance-friendly** than the current physical cash regime (which provides no issuance trail at all).

7.4 The CBDC Comparison

China's e-CNY, the European Central Bank's proposed digital euro, the Bank of Canada's CBDC research – all of these designs include either explicit or implicit surveillance capabilities. The central bank can see, freeze, expire, restrict, geofence, or retroactively examine any holder's balance.²

This is not an accident of the technology; it is a design goal. The pitch to governments is precisely that CBDCs give them visibility into civilian payment flows that physical cash denies them.

BUCK Cash is the opposite design. Visibility is at the edges (issuance, deposit), not the center (circulation). The issuer is identifiable, but only by selective disclosure with proper authorization (subpoena, court order, regulatory audit authority), not by everyday observation. The bearer-circulation phase is private the way cash is private.

A jurisdiction that adopts BUCK Cash gets **better compliance tools than physical cash provides today** without giving up **the privacy properties that make cash worth using**. This is a genuine Pareto improvement on the current regulatory trade-off and would, I believe, be politically defensible to even cash-skeptical authorities.

8 Comparison With Prior Art

This section places BUCK Cash in the lineage of digital cash designs from 1982 to the present, and identifies what each predecessor lacked.

8.1 Chaum's E-Cash (1982-1998)

David Chaum's blind-signature scheme⁴ introduced the core idea: an issuer (a bank) signs a token blinded by the holder so that the bank cannot link the issuance to the redemption. When the holder spends the token, the merchant verifies the signature without revealing who originally received it.

This is a beautiful design and operationally similar to BUCK Cash in spirit. Its failures were **institutional**, not cryptographic:

- Required a centralized issuer (the bank)
- Required online verification at spend time (no offline transferability)
- Two-sided market never bootstrapped (no merchants accepted, so no consumers held)
- DigiCash (the commercial implementation) went bankrupt in 1998

BUCK Cash has the same privacy properties without the centralized issuer (any identity-registered party can mint notes against their own collateral). Spend-time verification is on chain rather than via the issuer (so the issuer cannot selectively block redemptions). The two-sided market problem is mitigated by BUCK's existing on-chain ecosystem – BUCK already exists as a usable currency, so adopting BUCK Cash is a UX upgrade rather than a greenfield bootstrapping.

8.2 Brands' Offline Divisible E-Cash (1993)

Stefan Brands' scheme⁵ addressed the offline-spend problem: a divisible coin can be spent in pieces without contacting the issuer, but **double-spending mathematically reveals the spender's identity**.

This is the "deterrent rather than prevention" model. It works in theory but has two operational issues:

- The deterrent only deters if identity revelation has consequences (it relied on a legal/social enforcement layer external to the cryptography)
- The complexity of divisible-coin management was substantial and never produced a shipped product

BUCK Cash handles double-spend by **prevention** (global nullifier set), not deterrence. No legal layer is required. Divisibility is achieved by issuing multiple notes at burn time rather than by a single divisible coin.

8.3 Casascius Coins (2011-2013)

Mike Caldwell's Casascius coins³ were physical Bitcoin: a brass or silver coin with a holographic seal covering a private-key QR code. Snap the seal, scan the QR, the coin's Bitcoin balance was yours.

Casascius coins remain the closest existing analogue to BUCK Cash. What they lacked:

- Native integration with the underlying token (the Bitcoin protocol does not understand bearer-coin form; the coin is an aftermarket wrapper)
- Mode B targeting (all coins were bearer; no way to issue a coin claimable only by a specific recipient)
- Demurrage (a Casascius coin holds its full Bitcoin balance forever, unrelated to any economic mechanism)
- Programmable predicates (only one mode: bearer)
- Issuer was a single individual, who eventually stopped issuing in 2013 due to FinCEN money-transmitter concerns – an issuance bottleneck that BUCK Cash distributes to all identity-registered participants

BUCK Cash is, in essence, what Casascius would have been if Bitcoin had been designed with bearer-paper as a first-class instrument from the start, and if the underlying token had a demurrage and identity layer.

8.4 BIP-38 Paper Wallets

Encrypted Bitcoin private keys printed on paper, with a passphrase needed to spend. BIP-38 paper wallets are an aftermarket bearer instrument similar to Casascius but without the physical coin form factor.

Same limitations as Casascius (no native integration, no targeting, no demurrage, no predicates), plus the additional UX burden of remembering or transmitting the passphrase.

8.5 Zerocash / Zcash (2014-)

Ben-Sasson, Chiesa, Garman, Green, Miers, Tromer, and Virza's Zerocash⁷ gave us the first scalable on-chain commitment-and-nullifier UTXO model with full zero-knowledge. Zcash deployed it in 2016.

Zcash provides on-chain shielded transactions but does **not** extend the model to physical bearer instruments. All Zcash value lives on chain in shielded notes. No mechanism exists for printing a shielded Zcash note as a paper instrument that someone else could deposit later.

BUCK Cash uses Zerocash's commitment-and-nullifier primitive for its on-chain machinery and extends it with the printable-note layer. The contribution is in the extension, not the primitive.

8.6 Aztec / Zk.Money

Aztec's predicate-encrypted notes¹⁴ generalize Zerocash to private smart contract calls – a note can encode arbitrary state operated on by arbitrary predicates. This is the cryptographic ancestor of Mode C (programmable BUCK Notes).

Aztec does not have physical bearer instruments either. Its predicates are designed for on-chain DeFi applications (private DEX trades, private lending), not for paper notes. BUCK Cash applies the same predicate machinery to a different problem: specifying spend conditions for a printable bearer instrument.

8.7 Tornado Cash and Privacy Pools

Tornado Cash¹⁵ was the simplest possible commitment-and-nullifier mixer: fixed-denomination deposits in, anonymous withdrawals out. Sanctioned by OFAC in 2022 for money-laundering use.

Privacy Pools (Buterin, Schaefer, Tromer, et al., 2023)¹⁶ extends Tornado with per-spend "association sets" – a depositor can prove their deposit is in a non-incriminating subset, allowing compliance without identification.

Both are mixer designs, not cash-instrument designs. Their relevance to BUCK Cash is the lineage: BUCK's commitment pool is a Tornado-style global mixer, but with predicates and identity

¹⁴Williamson, Z., et al. "Aztec Protocol" (2018-). Ethereum L2 rollup providing general-purpose private smart contracts using PLONK proofs and predicate-encrypted notes. <https://aztec.network/>

¹⁵Pertsev, A., Semenov, R., Storm, R. "Tornado Cash Privacy Solution" (2019). Fixed-denomination commitment-and-nullifier mixer on Ethereum. Sanctioned by the US Office of Foreign Assets Control (OFAC) in August 2022 for alleged use in money laundering by North Korean state actors.

¹⁶Buterin, V., Schaefer, J., Tromer, E., et al. "Blockchain Privacy and Regulatory Compliance: Towards a Practical Equilibrium" (2023). Extends commitment-pool mixers with per-spend "association set" proofs, allowing depositors to demonstrate non-membership in incriminating subsets without revealing their specific deposit.

bound in. The selective-disclosure compliance story borrows from Privacy Pools' association-set ideas.

8.8 Comparison Summary

System	Bearer paper	Targeting	Demurrage	Predicates	Native	Decentralized	Year
Chaum E-Cash	No	No	No	No	-	No	1982
Brands Offline	No	No	No	No	-	No	1993
BIP-38 Paper Wallet	Yes	No	No	No	No	Partial	2010
Casascius Coin	Yes	No	No	No	No	No	2011
Zerocash / Zcash	No	No	No	No	-	Yes	2014
Tornado Cash	No	No	No	No	-	Yes	2019
Aztec	No	No	No	Yes	-	Yes	2020
Privacy Pools	No	No	No	Limited	-	Yes	2023
CBDCs (e-CNY, etc.)	No	Yes	Sometimes	No	Yes	No	2020+
BUCK Cash	Yes	Yes	Yes	Yes	Yes	Yes	2026?

The "Native" column captures whether the bearer-paper instrument is a first-class designed primitive of the system (as opposed to an aftermarket wrapper or absent entirely). BUCK Cash is the first design that fills every column. This is not because the prior art was incompetent; it is because the four required pieces (commitment-pool primitive, identity layer, demurrage, jurisdictional reserve) finally converged in one system.

9 Comparison: Cash, CBDC, BUCK Cash

A more direct comparison aimed at a non-cryptographer reader: how does BUCK Cash stack up against physical cash and central bank digital currencies?

Property	Physical Cash	CBDC (typical)	BUCK Cash
PRIVACY			
Holder's identity hidden	Yes	No (issuer sees all)	Yes (during circulation)
Transaction graph hidden	Yes	No	Yes
Issuer's identity hidden	Yes (notes are fungible)	No	No (selective disclosure)
FINALITY			
Settlement is instant	Yes	Yes	Yes (chain finality)
Reversible by issuer	No	Yes (CBDC freeze)	No
Reversible by holder	No	Yes (chargeback)	No
INFRASTRUCTURE			
Works without internet	Yes	No	Yes (transfer); needs net for deposit
Works without battery	Yes	No	Yes (paper); needs power for deposit
Works without bank account	Yes	No	Yes
Cross-border in your pocket	Yes (with declaration)	No (geofenced)	Yes
DURABILITY			
Survives infrastructure failure	Yes	No	Yes (paper); chain on recovery
Counterfeit-resistant	Partial (printed security)	Yes	Yes (cryptographic)
Recoverable if lost	No	Yes (issuer can re-issue)	No (but demurrage returns value to system)
ECONOMIC BEHAVIOR			
Lost notes extracted forever	Yes	No	No (returned via demurrage)
Inflation tracks issuer policy	Yes	Yes	Yes (tracks BuckK)
Holds value over decades	No (inflation)	No (inflation)	No (demurrage 2%/yr)
Holds value over years	Approximately	Approximately	Yes (with demurrage discount)
PROGRAMMABILITY			
Time-locked spends	No	Possible	Yes
Conditional payouts	No	Possible	Yes
Multi-signature spending	No	Possible	Yes
Issuer can restrict spending	No	Yes	No (issuer relinquishes control at print)
COMPLIANCE			
Issuance is auditable	No	Yes	Yes
Circulation is auditable	No	Yes	No (private)
Deposit is auditable	Partial (CTR rules)	Yes	Yes
Selective disclosure available	No (no records)	No (full disclosure)	Yes

The dominant insight from this table: **BUCK Cash combines the privacy properties of physical cash with the compliance and durability properties of CBDC, while preserving infrastructure-independence that CBDC abandons.**

The single property where physical cash is genuinely irreplaceable – "works without internet for deposits" – is moderated for BUCK Cash but not eliminated. Holders can transfer BUCK Notes hand-to-hand offline, indefinitely; they need internet only when they want to convert the bearer instrument to an on-chain balance. In a disaster scenario, BUCK Notes circulate as cash for the duration of the outage, then deposit normally when the chain becomes reachable again.

10 Implementation Path

This section sketches what would actually need to be built to ship BUCK Cash.

10.1 Phase 1: On-Chain Note Pool

The prerequisite for everything else is the global commitment pool described in Wormhole, "Global Commitment Pool" section. This is a v2 of the BUCK contract:

- Add a Merkle tree of note commitments

- Add a global nullifier set
- Add a SNARK verifier for the standard "spend a note" predicate (Mode A bearer)
- Add a `burn_to_notes` entry point that takes a value and a vector of commitments, appending to the tree

Engineering scope: roughly the size of an Aztec-style note pool. Substantial but well-understood; multiple production reference implementations exist (Aztec, Zcash Sapling, Tornado Cash variants).

10.2 Phase 2: Mode A Bearer Notes

With the note pool in place, Mode A is essentially "Tornado Cash with a paper output." Build:

- A wallet app that prints QR codes after a successful `burn_to_notes`
- A wallet app that scans QR codes and submits spend SNARKs
- A printer driver / template library for note layouts
- A scratch-off / hologram seal vendor relationship for high-trust deployments

Engineering scope: a smartphone app and a small print-shop integration. Not large.

10.3 Phase 3: Mode B Targeted Notes

Add the stealth-address derivation $pk^* = H(s) \cdot pk_{\text{recipient}}$ and the corresponding spend predicate. This re-uses the Chaum-Pedersen machinery from the Identity protocol.

Engineering scope: a SNARK circuit extension and corresponding wallet-app UX. Modest.

10.4 Phase 4: Mode C Programmable Notes

Add a circuit-compiler integration so that note printers can supply arbitrary predicates at issuance time. Aztec's Noir compiler is the model here – a domain-specific language for SNARK predicates that compiles to verifiable circuits.

Engineering scope: substantial, but draws heavily on existing Aztec/Noir tooling. Could be deferred until after Phases 1-3 prove out the core concept.

10.5 Phase 5: Off-Chain Re-Targeting (Optional)

Proxy re-encryption (e.g., Umbral¹⁷) over Mode B notes would permit off-chain transfer between identities without consuming the nullifier. This is the "Lightning Network for bearer instruments" extension.

Engineering scope: requires a network of proxy nodes (the Identity Guardians could play this role) and a re-encryption circuit. Defer until Phases 1-4 demonstrate demand.

¹⁷Nunez, D. "Umbral: A Threshold Proxy Re-Encryption Scheme" (2018). NuCypher's threshold proxy re-encryption scheme based on the BBS98 PRE construction. Allows a delegator to grant a delegatee decryption rights to a ciphertext via a network of N proxies, no proper subset of which can compromise the underlying plaintext.

10.6 Regulatory and Legal Engagement

In parallel with engineering, the legal and regulatory work:

- Confirm that BUCK Cash falls within Alberta's provincial jurisdiction over property and civil rights (s 92(13)) – the analysis in the Legal Foundation document covers the underlying primitive; bearer notes are an extension that may warrant additional treatment
- Engage FINTRAC on the issuance-side reporting model
- Engage the Office of the Superintendent of Financial Institutions (OSFI) on reserve treatment
- Coordinate with Alberta Treasury Branches (ATB) as a candidate first issuer

11 The Larger Picture

I want to close by stepping back from the technical detail and considering what BUCK Cash is, in the broader arc of this series.

The Alberta Buck began as an answer to a specific question: how can civilians access the third monetary element – Claim money – that banks currently monopolize? (The Missing Monetary Element develops the framework.)

The system that emerged – BUCK_CREDIT, BUCK, BuckK, demurrage, Jubilee, the Identity Fountain, the Wormhole – is a complete monetary infrastructure that operates on chain, with smartphone-mediated user access. It is sufficient for an all-digital economy.

But Alberta is not an all-digital economy, and neither is anywhere else. Cash – real, physical, hand-to-hand, no-infrastructure-required cash – remains essential for billions of people, in normal times and especially in abnormal ones. A monetary system that does not have a cash form excludes the populations who need it most.

BUCK Cash extends Alberta Buck into the cash-using world without compromising any of the on-chain properties. The same demurrage, the same identity, the same reserve, the same Jubilee – in a form factor that works at a farmers' market, in a remote community without reliable cellular coverage, in a power outage, across a border, in a wallet, under a mattress, in an envelope to a relative abroad.

It is, I think, the missing layer that turns Alberta Buck from a sophisticated financial protocol into something the unbanked elderly can use as easily as the crypto-literate young. And it is, I think, the design that finally answers the question Chaum was asking in 1982: what does digital cash look like, when you take "cash" seriously?

The answer, forty years later: it looks like a piece of paper with a QR code, a hologram, a denomination, an issuer mark, and a value that decays at the same rate as your bank balance does. Indistinguishable from cash to the user; cryptographically distinguishable from anything counterfeit; integrated with a complete monetary system that doesn't require a central bank to function.

Alberta has, uniquely in the world, all four pieces required to build this. Whether to assemble them is, as ever, the only question that matters.